Least-square: First-order Linear System

Consider a linear dynamical equation

$$x(t+1) = ax(t) + bu(t), \quad x(0) = x_0$$

where $a, b \in \mathbf{R}$ with $|a| < 1$. We randomize $u(t) \in \{-1, 1\}$ to the system and due to an additive corrupted noise, the measurement y is not exactly given above. Given $\{u(t), x(t)\}_{t=1}^{N}$, formulate two least-squares problem to estimate $\hat{y}$.

You can collect the dataset by assuming the ground-truth parameter: a, b, setting an initial condition x_0, specifing the number of time points, N, and calling the following function from such values. It returns t, y, x, and plot the graphs.

```
import numpy as np
import matplotlib.pyplot as plt

def sim_discrete_firstorder(a, b, x0, N):
    np.random.seed(0)
    x = np.zeros(N)
    x[0] = x0
    u = np.sign(np.random.randn(N))

    for k in range(N-1):
        x[k+1] = a*x[k]+b*u[k] + 0.1*np.random.randn()

    t = np.arange(N)

    return x, u, t

# then plot the graph
```

Problem

1. Consider the dynamical equation and use least-squares to estimate a, b directly. Write down the linear system $y = X\beta$, and write python code to solve $\hat{\beta} = (a, b)$. Then compute $\hat{y}$. Your function has the following format. The output are $y, X, \hat{\beta} \in \mathbf{R}^2$ and $\hat{y}$.

   ```
   import numpy as np

   def LSestimate_1storder(x,u):
       ...
       return y, X, betahat, yhat
   ```

2. Plot $\hat{y}$ from the linear regression model and compare with the actual measurement $x(t)$. Submit this part to paper homework.

Implementation

Goal: Complete the code as the problem state and submit to the Grader.

```
import numpy as np

# Ground-truth data generation
def sim_discrete_firstorder(a, b, y0, N):
    np.random.seed(0)
    y = np.zeros(N)
    y[0] = y0
    u = np.sign(np.random.randn(N))

    for k in range(N-1):
        y[k+1] = a*y[k]+b*u[k] + 0.1*np.random.randn()

    t = np.arange(N)

    return y, u, t

def LSestimate_1storder(x, u):
    # TODO: ...
    ...
    return Y, X, betahat, yhat

if __name__ == "__main__":
    # please import matplotlib here to avoid issues in grader environment
    import matplotlib.pyplot as plt
    # ---------- Generate dataset ----------
    N = 100
    a = -0.8
    b = 0.3
    y0 = 0
    y, u, t = sim_discrete_firstorder(a, b, y0, N)

    plt.figure(figsize=(8,6))
    plt.subplot(2, 1, 1)
    plt.step(t,y, where = 'post',label= r"Output $y(t)$")
    plt.xlabel('Time')
    plt.legend()

    plt.subplot(2, 1, 2)
    plt.step(t,u,'r', where = 'post',label= r"Input $u(t)$")
    plt.legend()
    plt.xlabel('Time')
    plt.show()

    # --------- Perform Least-square estimation -----
    betahat, yhat = LSestimate_1storder(y, u)

    # --------- Plot of prediction vs Ground-truth ----
    # TODO: put your own plot here
    # Plot does not affect the score on Grader but highly recommend to practice.
    ...
```